Build your own free SOC Lab with Splunk

minder-security.ghost.io/get-started-with-splunk

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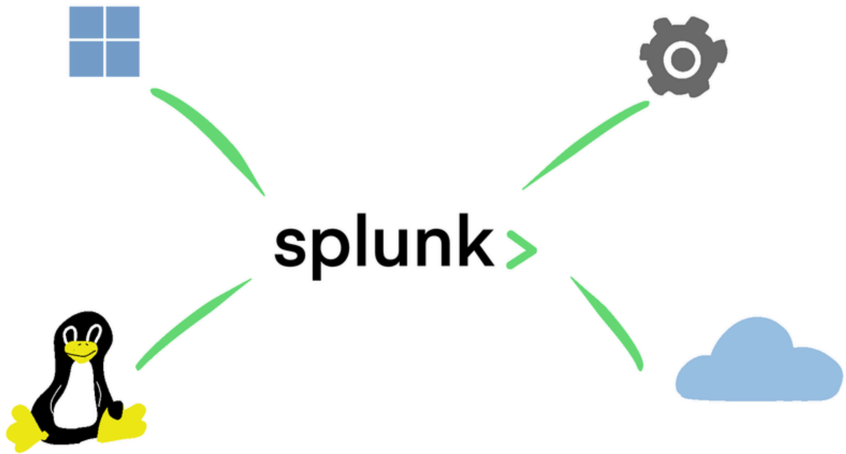


The term Splunk is associated with software solutions widely used in the fields of data analysis. In this tutorial, I will use what I learned during my time as a SOC Analyst to showcase how you can build your own SOC at home for free.

Lab Setup Tutorial

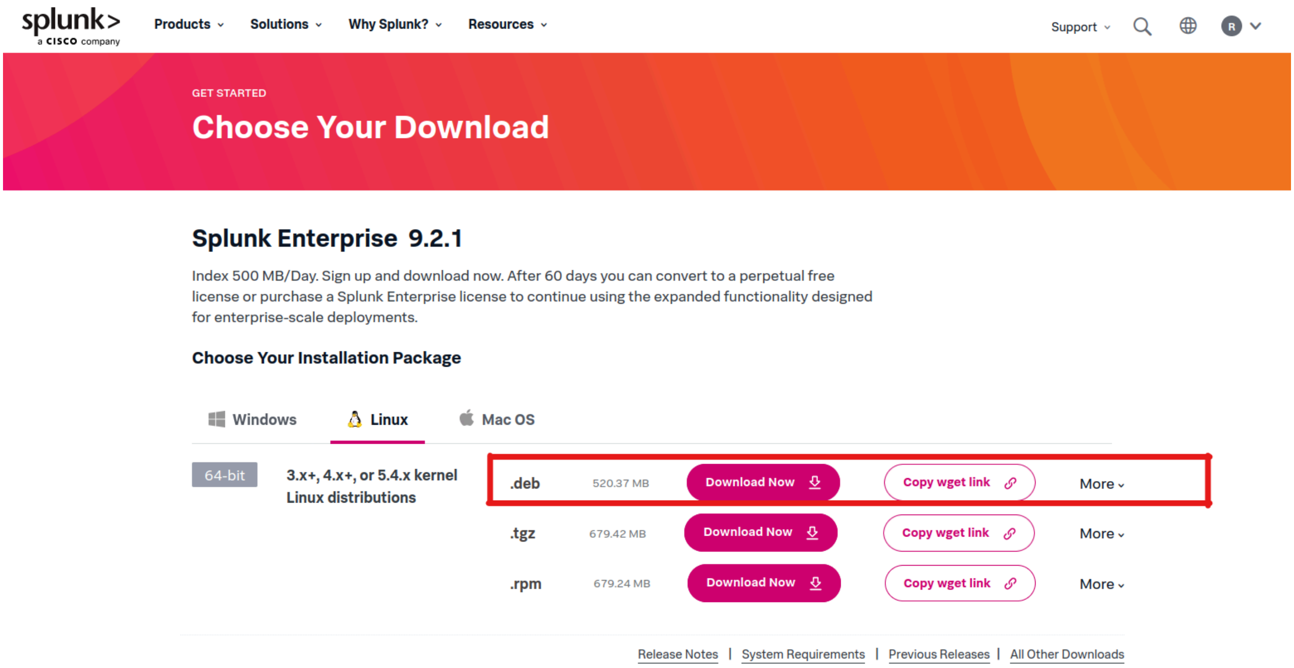
Our simplified Architecture will consist of the following components:

- **Ubuntu virtual machine** running Splunk Enterprise.
- **Windows 11 Enterprise (Evaluation) virtual machine** running Splunk Forwarder, which forwards data to our Ubuntu Splunk Enterprise Server.

1 - Installing Splunk Enterprise on Ubuntu

Scroll further down for a Cheat-Sheet of most useful Splunk CLI commands
After you setup your VM's, it's time to **Install Splunk Enterprise on our Ubuntu Virtual Machine**.

1. First, you need to create a free account to download the free trial version of Splunk Enterprise. I have decided to **download the .deb package**:



2. After the download was completed, **I installed the package** using the following command in the CLI:

Make sure **curl** is installed on your system!

```
sudo dpkg -i ./splunk<version>.deb
```

You can verify the installation using the following commands:

```
dpkg --status splunk
dpkg --list
```

3. Then **start Splunk Enterprise for the first time** (note that the file path might vary depending on where you installed Splunk):

```
sudo /opt/splunk/bin/splunk start
```

4. You will be prompted to **accept the general terms**. Just press "**q**", then press "**y**" to accept the terms followed by pressing Enter:

```
SPLUNK GENERAL TERMS

Last Updated: August 12, 2021

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Do you agree with this license? [y/n]: y
```
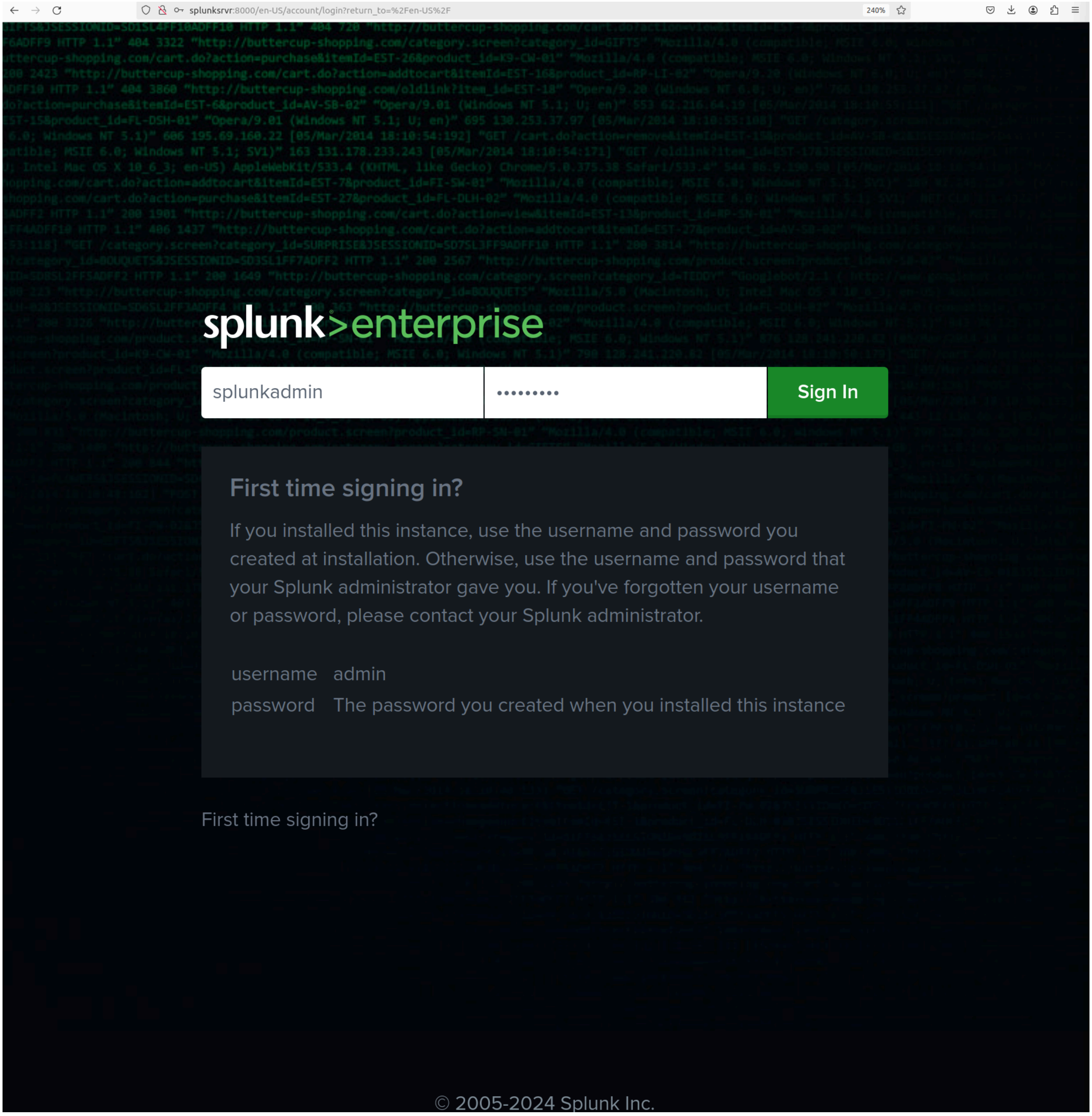
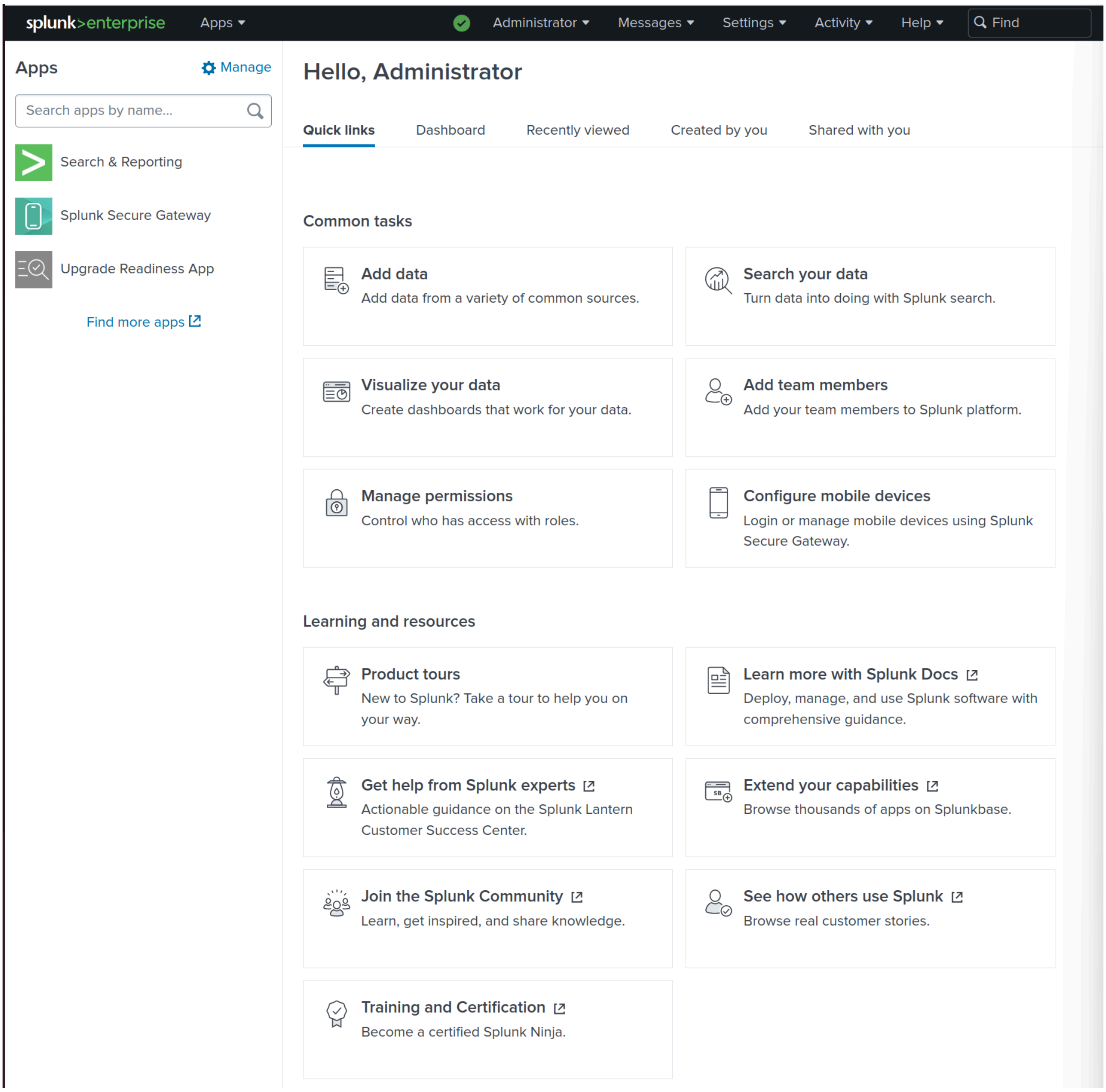

5. Then **setup the credentials** you want to use for your **Splunk Admin Account**. After you entered your credentials, Splunk Enterprise will be installed and you will be given a **Web-Interface URL**:

```
Starting splunk server daemon (splunkd)...
Generating a RSA private key
..........+++++
.................................................................................+++++
writing new private key to 'privKeySecure.pem'
-----
Signature ok
subject=/CN=splunksrvr/O=SplunkUser
Getting CA Private Key
writing RSA key
PYTHONHTTPSVERIFY is set to 0 in splunk-launch.conf disabling certificate validation for the httplib and u
rllib libraries shipped with the embedded Python interpreter; must be set to "1" for increased security
Done


Waiting for web server at http://127.0.0.1:8000 to be available............ Done


If you get stuck, we're here to help.
Look for answers here: http://docs.splunk.com

The Splunk web interface is at http://splunksrvr:8000
```

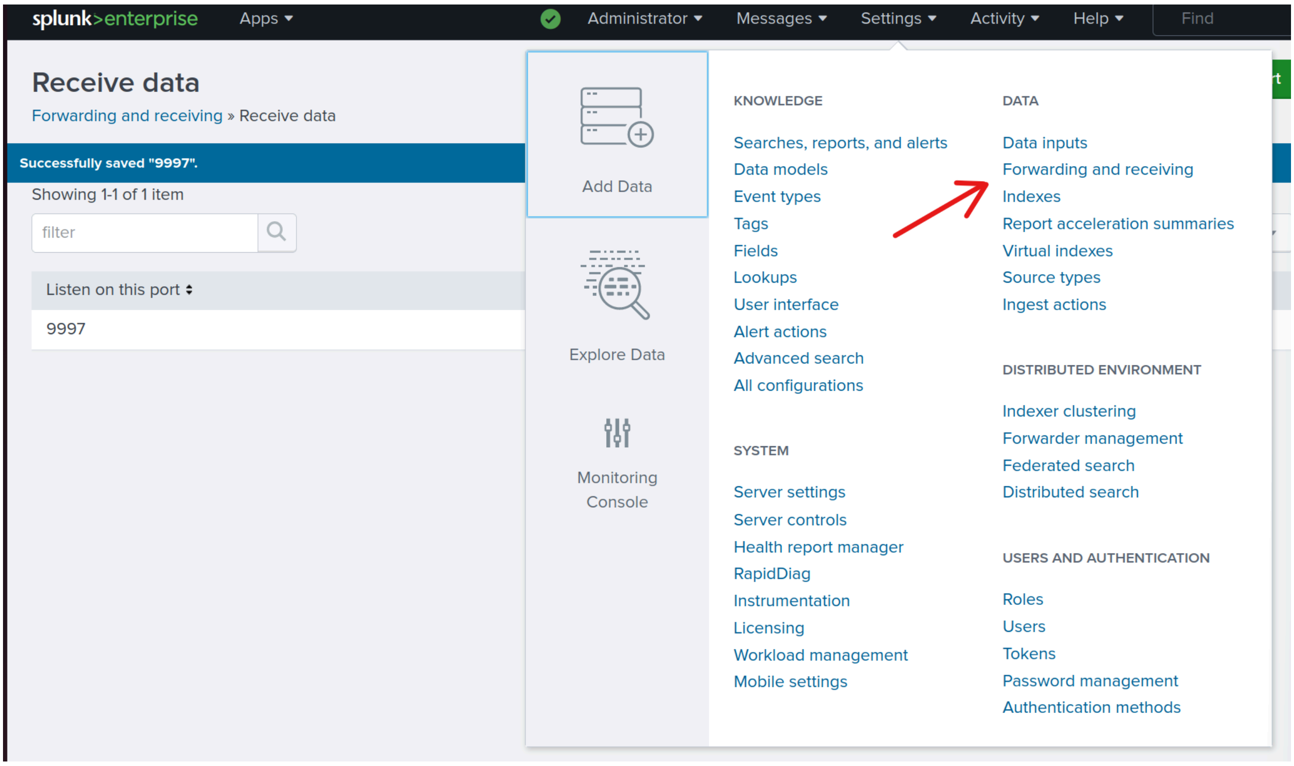
6. **Open the URL inside your Webbrowser**. You will be greeted by a login-page looking similar to this one. Use your credentials so authenticate:



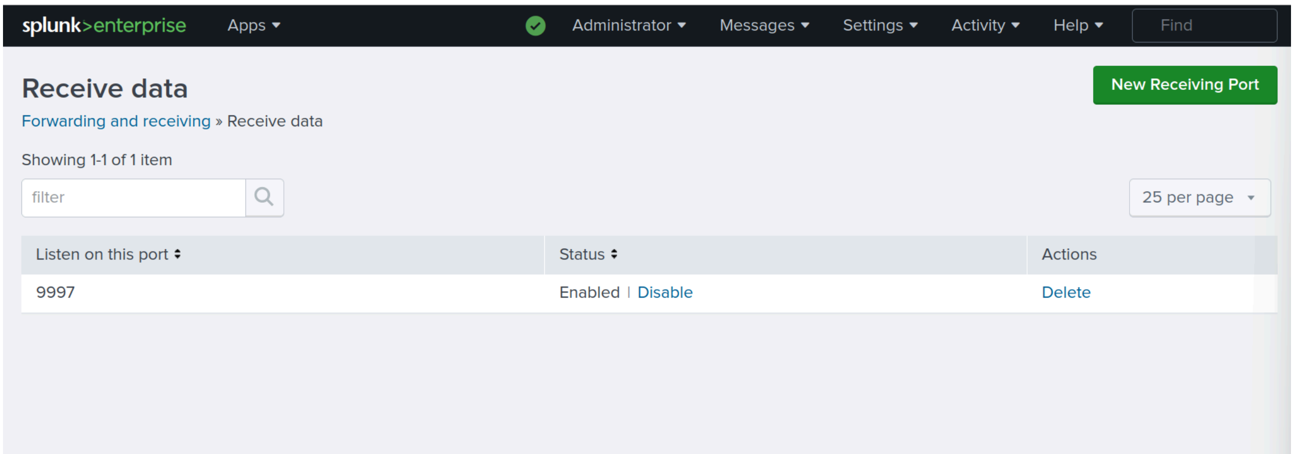
7. After authenticating, you will be greeted by the **Splunk Enterprise Homepage**:



8. **Configure Splunk Enterprise to receive Data from Forwarders** by going into Settings>Forwarding and receiving:



9. Click on **Configure receiving** > **New Receiving Port**. Enter the port you wish to use to receive Data from Forwarders. I chose the default Port 9997. Click on **Save**, then you should see the following configuration on your page:



10. Verify that your Listening Port is enabled, then check the IP-Address of your Splunk Server using the following command:

```
ip addr
```

```
xhor@splunksrvr:~/Downloads$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:bb:29:e7 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 192.168.137.148/24 brd 192.168.137.255 scope global dynamic noprefixroute ens33
       valid_lft 601055sec preferred_lft 601055sec
    inet6 fe80::4065:86b2:1269:9fdb/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
```

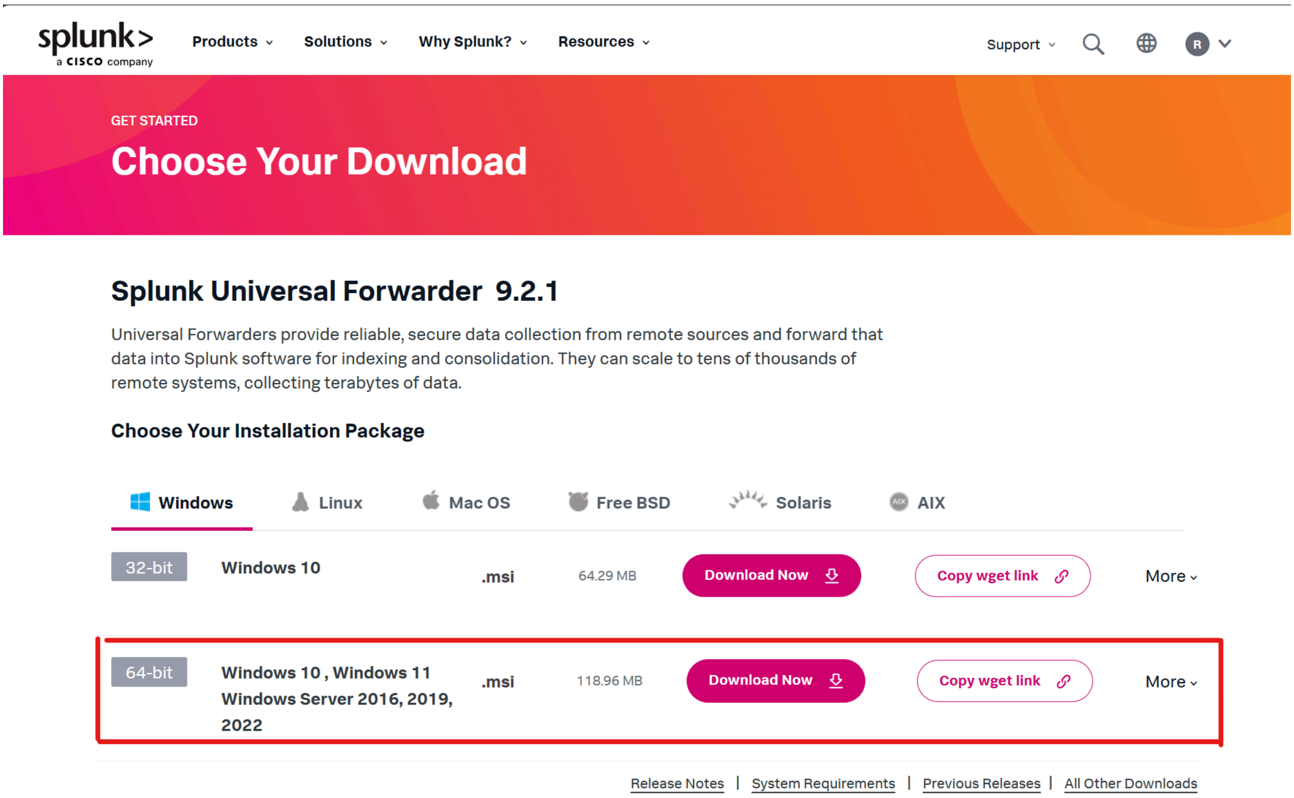
Note down the IP-Address your server uses.

I'm using a Bridged Network Setup for my Ubuntu VM

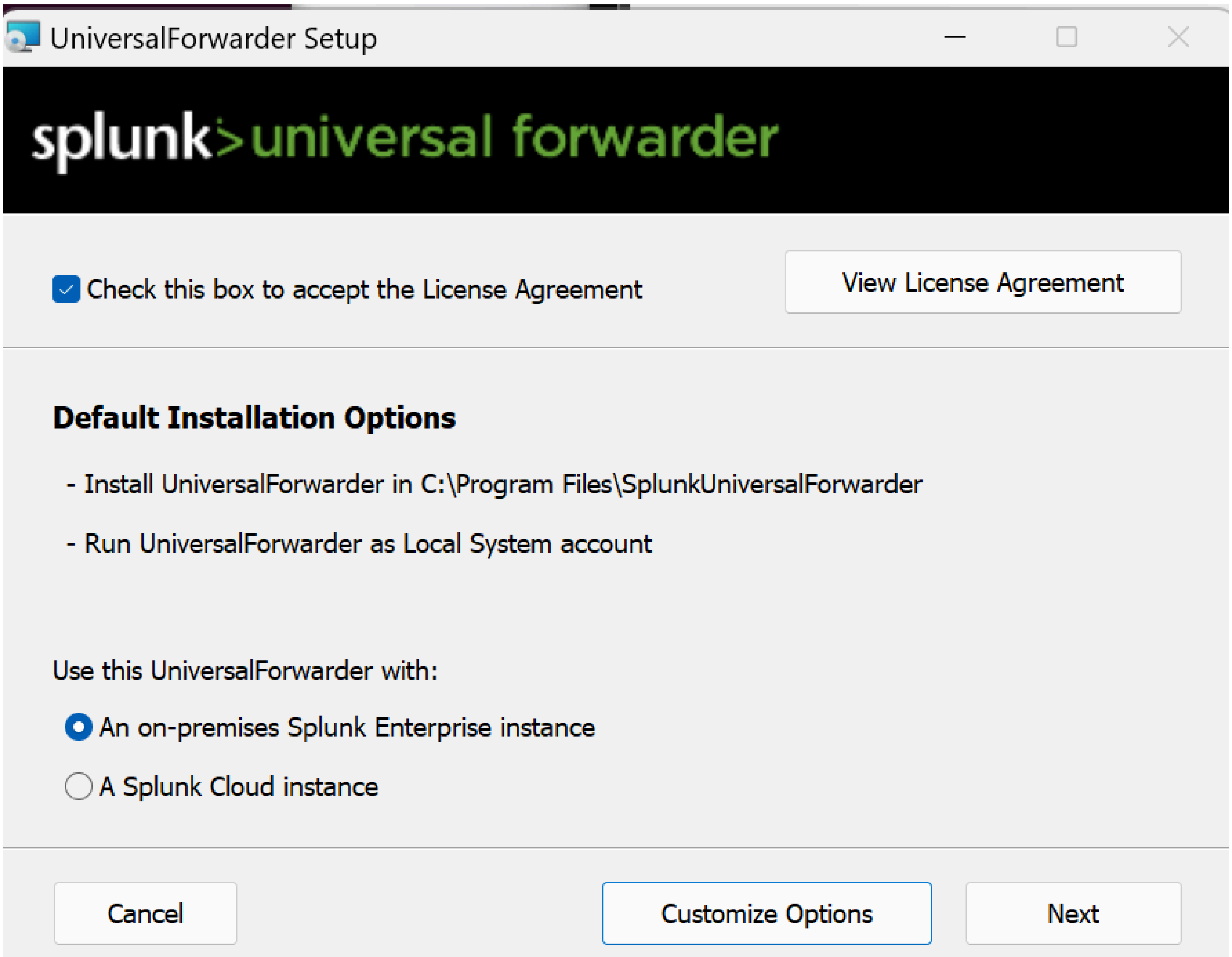
2 - Forwarding Data from Windows Hosts

After Splunk Enterprise has been installed on our Ubuntu Server, it's time to forward data to it. We will do so using **Splunk Universal Forwarder**. Follow these steps:

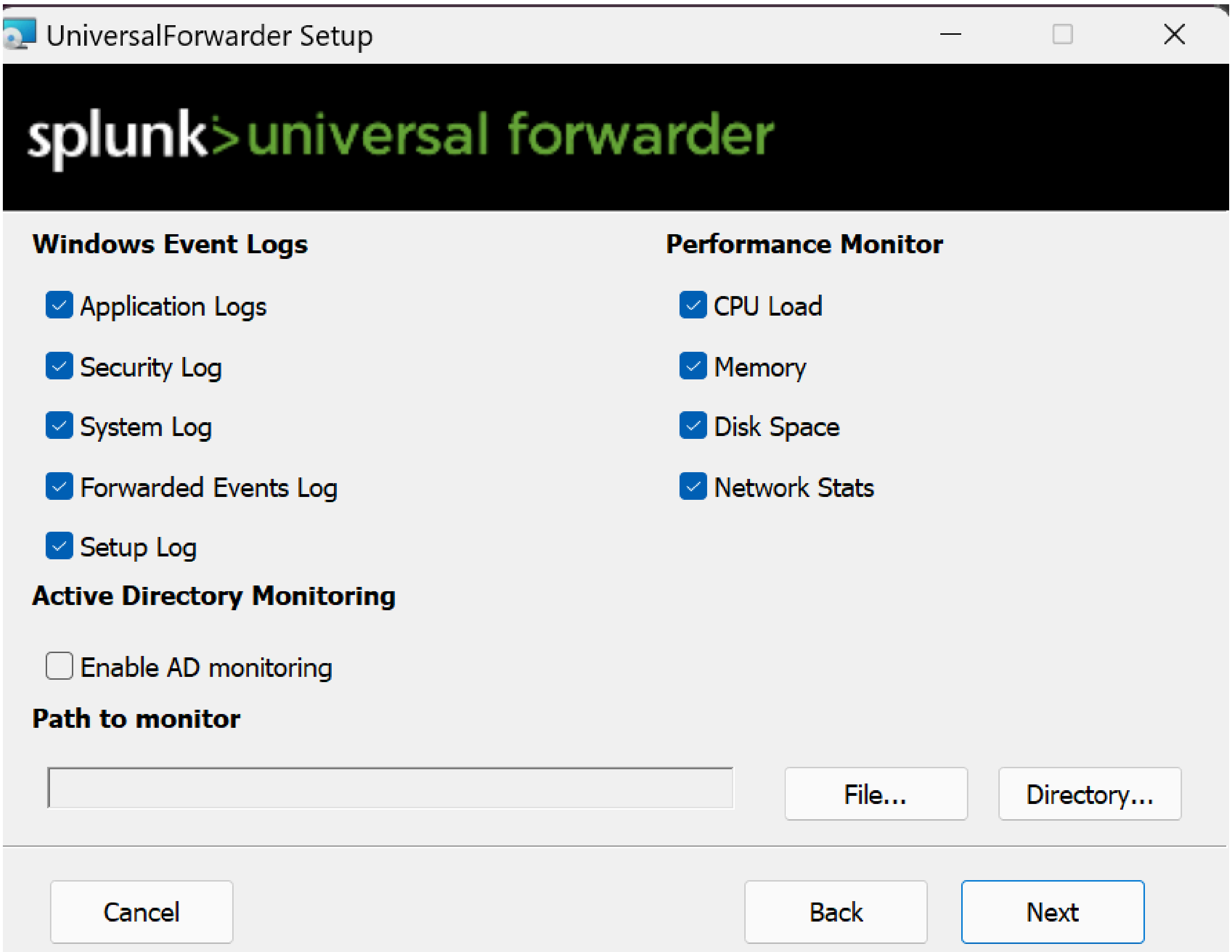
1. **Download the Splunk Universal Forwarder onto the Windows Host**. Again, do so using your free account. I installed the 64-Bit Version:



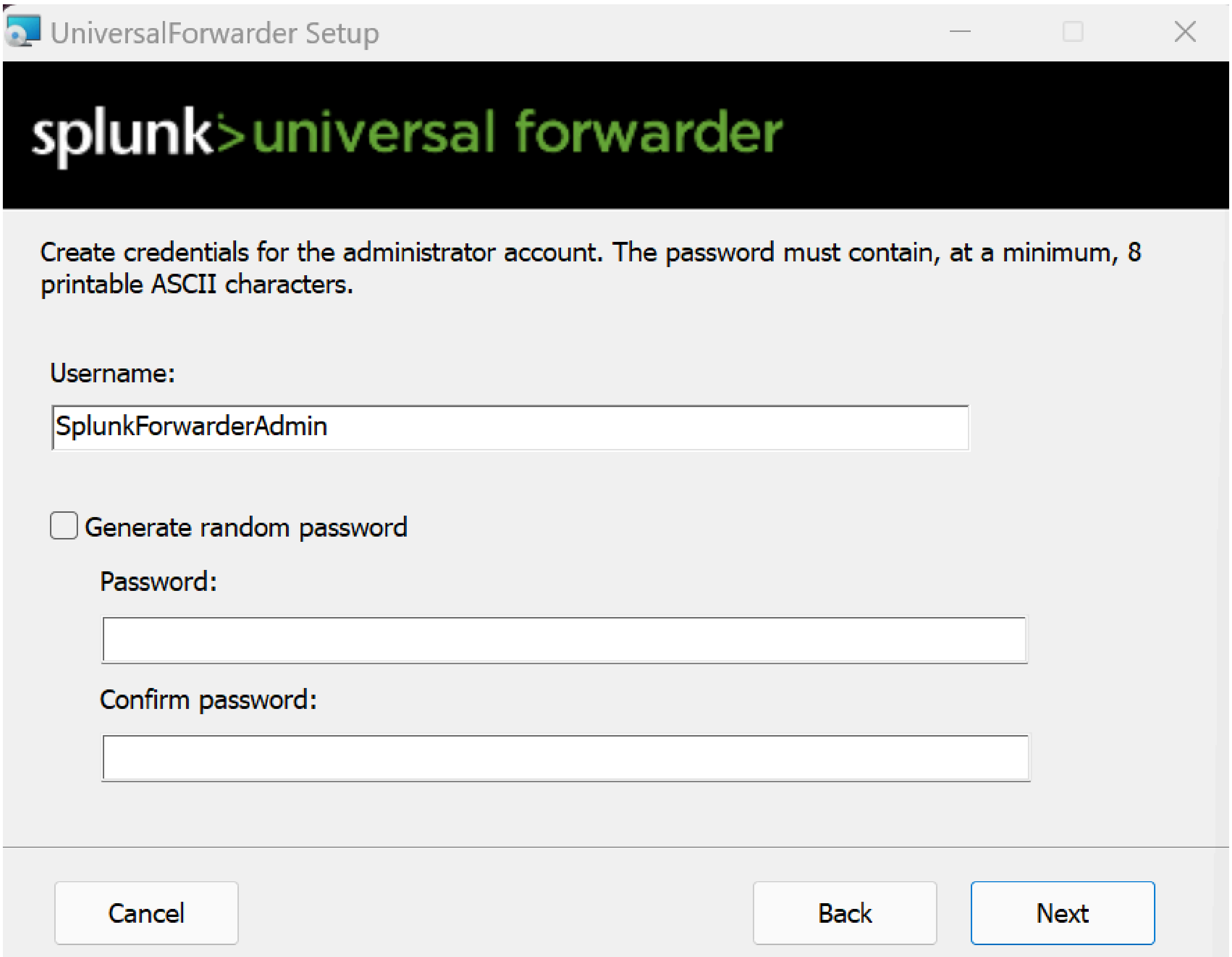
2. **Execute the downloaded .msi File** on your Windows host. Check the License Agreement Box. Then click **Customize Options:**



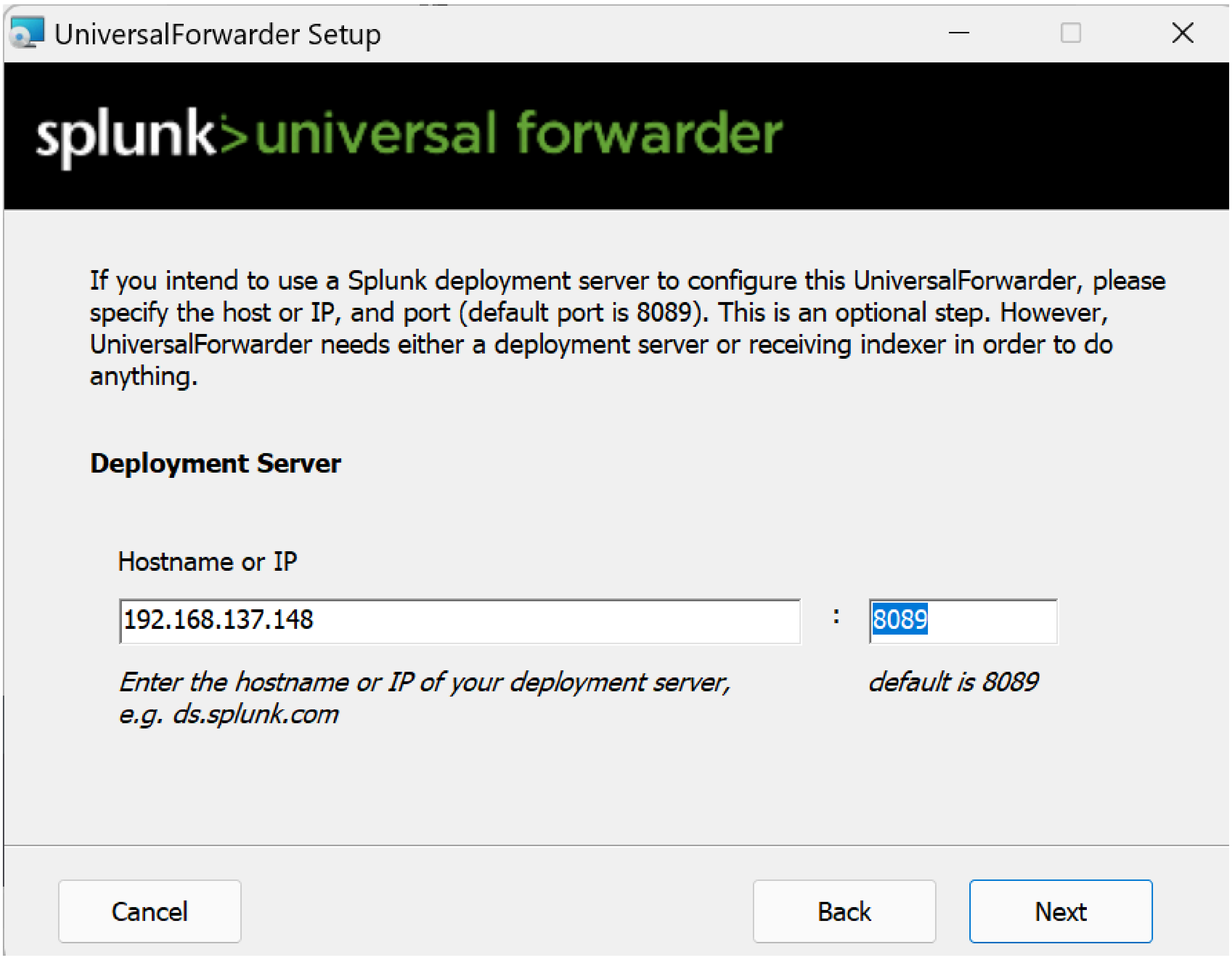
3. Click **Next** until you get to the page shown in the picture below. **Select all the data sources you want to collect**. I chose everything, just for the sake of curiosity. Then click on **Next**:



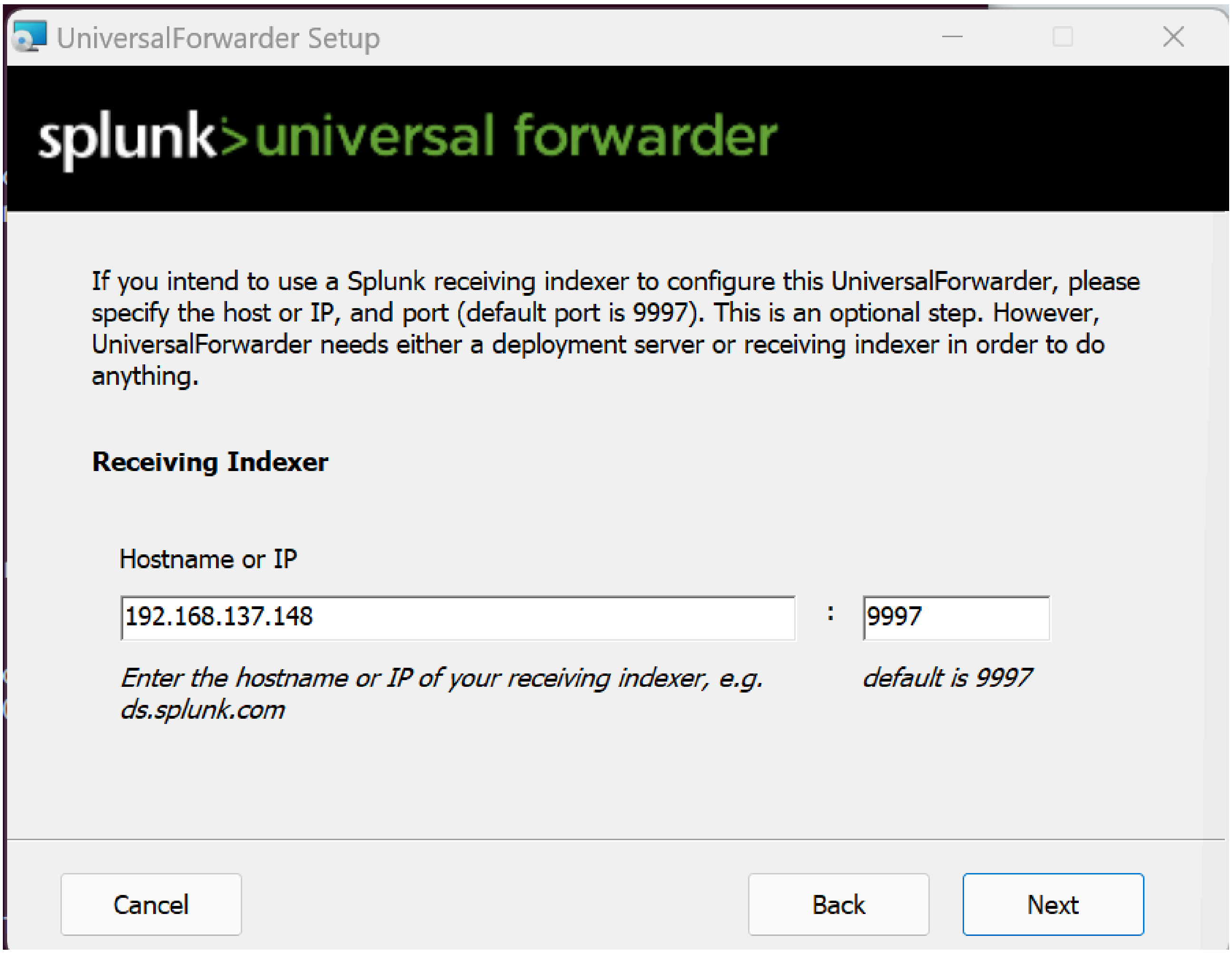
4. **Create an administrator account for the forwarder.** Click **Next** after you're done:



5. In this step, we can configure a deployment server, which is useful when you have multiple hosts to manage. Enter the **hostname/IP of your Splunk Enterprise Server** and the **Port** (default if you haven't configured anything else). Then proceed by **clicking Next**.



6. Now **enter the hostname/IP of your Splunk Enterprise Server**, as well as **the Port you configured for receiving Data:**

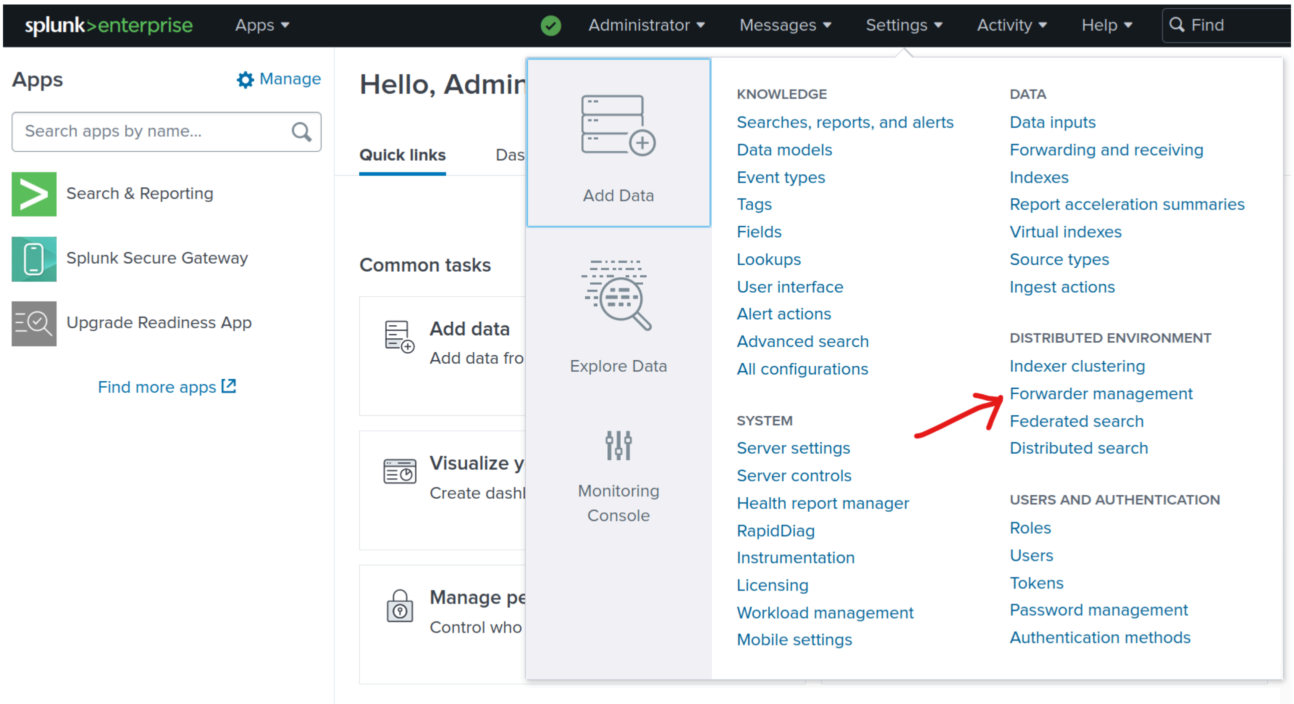


7. Click on **Install**, then **Finish**.

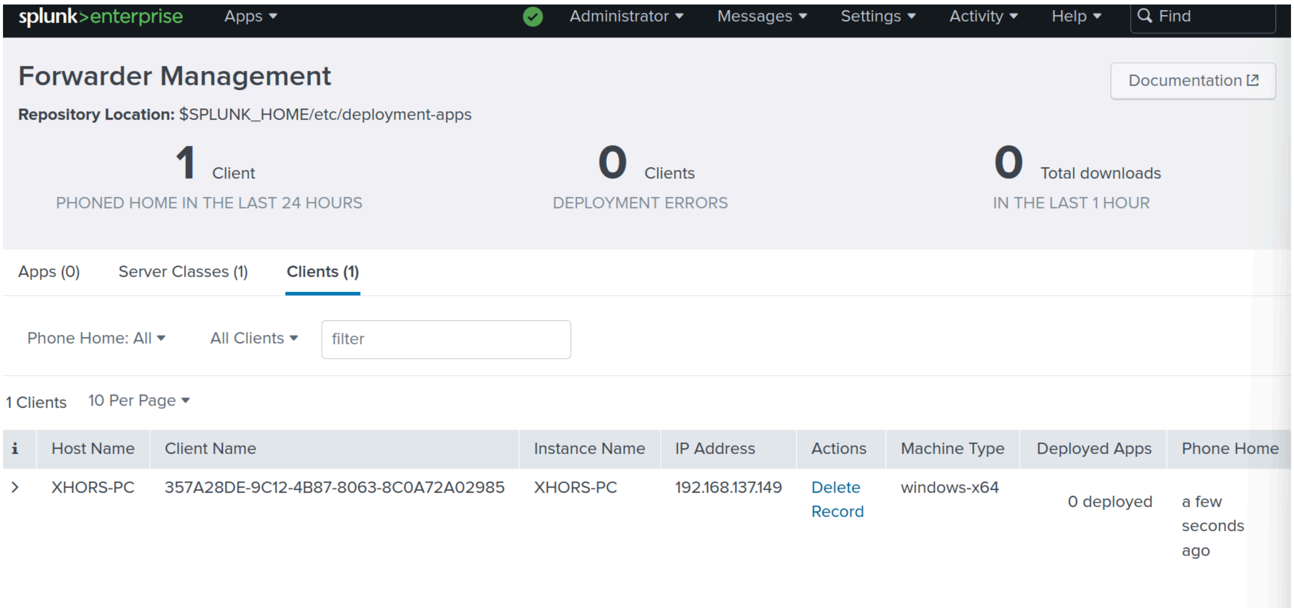
To uninstall your configuration, stop the service from running and uninstall Splunk Universal Forwarder in your Windows App-Settings.

3 - Testing your Setup

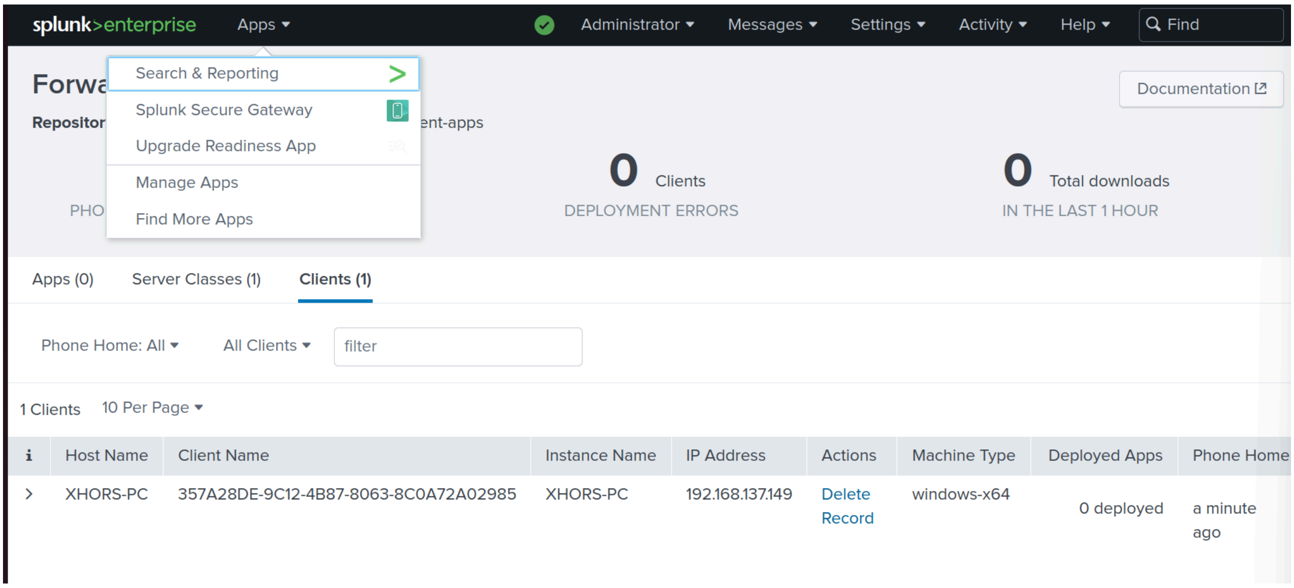
Now you have successfully configured the client to forward Data to our Splunk Enterprise Server running on Ubuntu Linux. If you configured the **Deployment Server**, you can verify that the Forwarder has connected to the Server by opening **Settings > Forwarder Management:**



Then you should be able to see your onboarded Device:



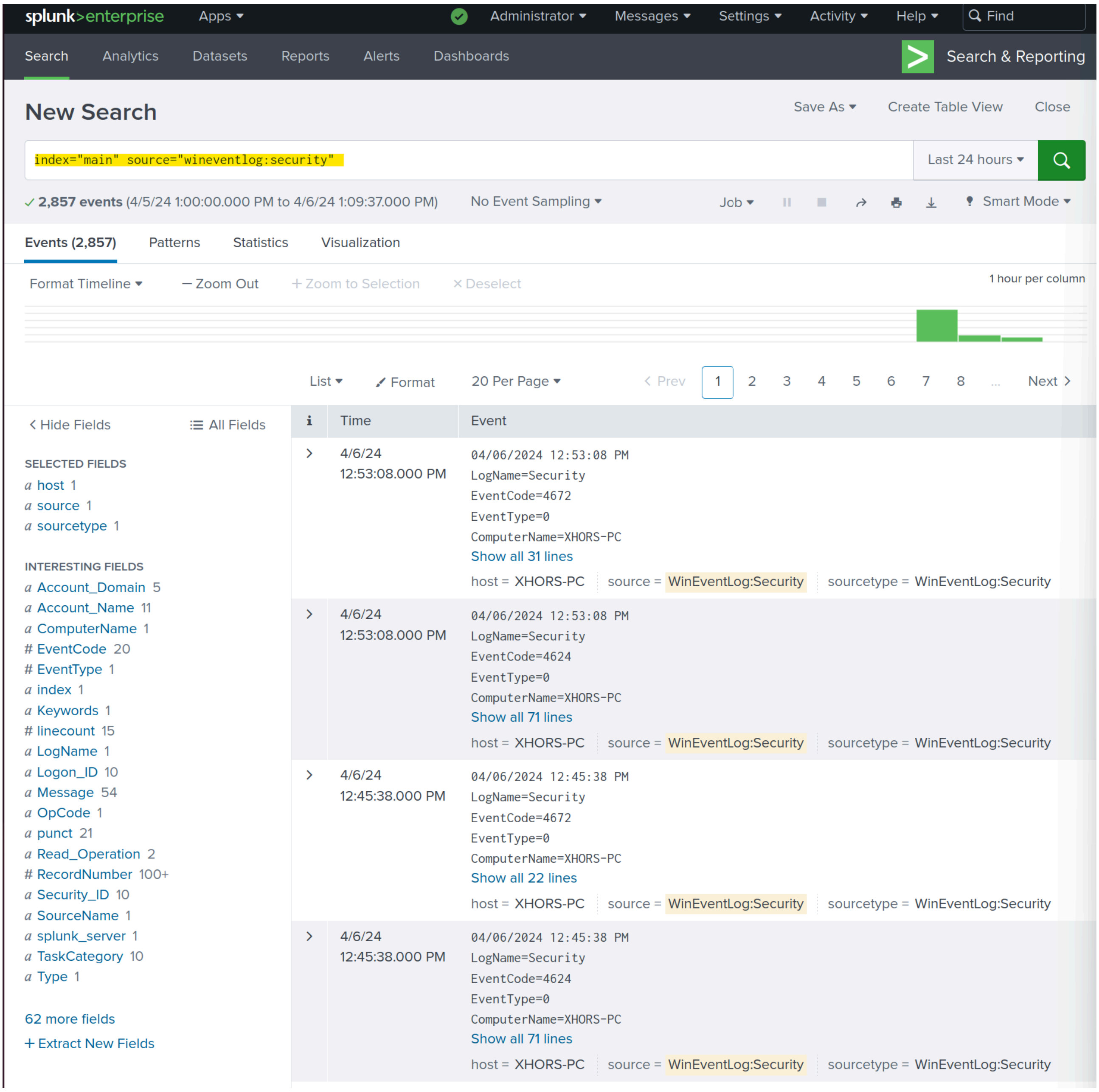
To run Splunk Processing Language (SPL) Queries, you can go to **Apps > Search & Reporting** and start querying data:



You can start by typing the following line into the search bar:

```
index="main" source="wineventlog:security"
```

This line will return all logs collected from our Windows Security Event Logs. You can start to play around with the GUI and try out some SPL queries:



Now you have everything you need to get started with Splunk Enterprise!

Thank you for reading my blog post and I hope I could help you build your own SOC Lab!

Additional Info: Splunk CLI Cheat Sheet

```
Starting, Stopping, Restarting and checking Status of the Splunk Enterprise Server
==================================================================================

splunk start

splunk stop

splunk restart

splunk status
```

```
Add a Test-Event to the Splunk Index
===================================================================================

splunk add oneshot



Search Data in the Splunk Index
===================================================================================

splunk search <keyword>



More Information
===================================================================================

splunk help
```